

Discussion on the influence of segmentation quality

Semantic segmentation plays a critical role in our method, as both the semantic features and the fuzzy rules are derived from the image segments. Although segmentation quality may affect explanation performance, our current framework can address this issue effectively. In the following, we discuss this concern from three complementary perspectives:

- the mechanisms within the framework that alleviate the impact of inaccurate segmentation on the explainer’s performance,
- the empirical influence of inaccurate segmentation on explanation results,
- the possible solutions within the proposed framework for handling inaccurate segmentation.

Two additional case studies will be present next to demonstrate these aspects in practice.

(1) Mechanisms within the framework for handling segmentation uUncertainty

From a theoretical perspective, the proposed GT2-HFMAE-I method intrinsically handles segmentation uncertainty at two levels.

First, the explainer uses the primary membership function (PMF) to capture uncertainty from the image segmentation, thereby reducing the sensitivity of the explanations to segmentation quality. Unlike approaches that directly map superpixels to semantic segments in a crisp manner, GT2-HFMAE-I models the degree of membership of each superpixel to all semantic segments through the PMF formulation (Eq. 5 in the manuscript). Furthermore, by sampling the threshold parameter ε in Eq. 6, the framework models the footprint of uncertainty (FOU) corresponding to these mappings. This enables the explainer to represent ambiguous or overlapping boundaries where a single superpixel may partially belong to multiple semantic components. Consequently, the quality of the semantic segmentation is not a decisive factor for the explanation, as the explainer primarily learns the degree associations between superpixels and semantic segments rather than a fixed, discrete segmentation map. In this way, the uncertainty modeling embedded in PMF provides an inherent robustness against inaccurate or noisy segmentation results.

Second, the semantic axis of the biaxial hierarchical framework can also alleviate the potential influence of segmentation quality on the resulting explanations. For complex or cluttered images, users can select from different segmentation algorithms, combine multiple segmentation results, or construct explanations at varying levels of granularity along the semantic axis from simplified to refined levels. Since the segmentation process is the first step of the GT2-HFMAE-I algorithm (see Fig. 3 in the manuscript), preceding dataset sampling and explainer training, such adjustments do not introduce additional computational overhead. This modular design enables users to adaptively improve or adjust segmentation quality without training new explainer.

(2) Experiment analysis: influence of inaccurate semantic segmentation on explanations

To further investigate the effect of inaccurate semantic segmentation on the quality of explanations, we conducted two representative case studies of DenseNet-121 [1] fine-tuned by CUB-200-2011 dataset [2]. These examples illustrate how different types of segmentation inaccuracy influence the explanatory results and how the proposed framework alleviates such effects.

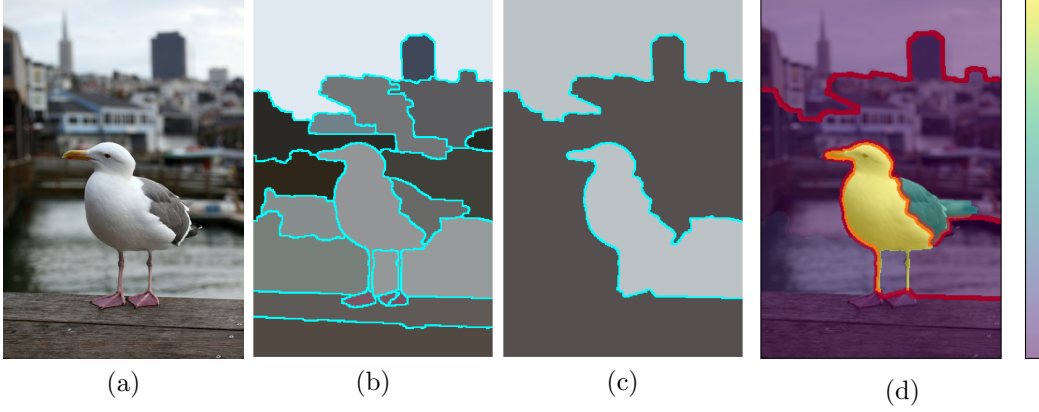


Figure 1: Influence of inaccurate semantic segmentation on Instance Western_Gull_0087_54193 in CUB-200-2011 dataset [2].

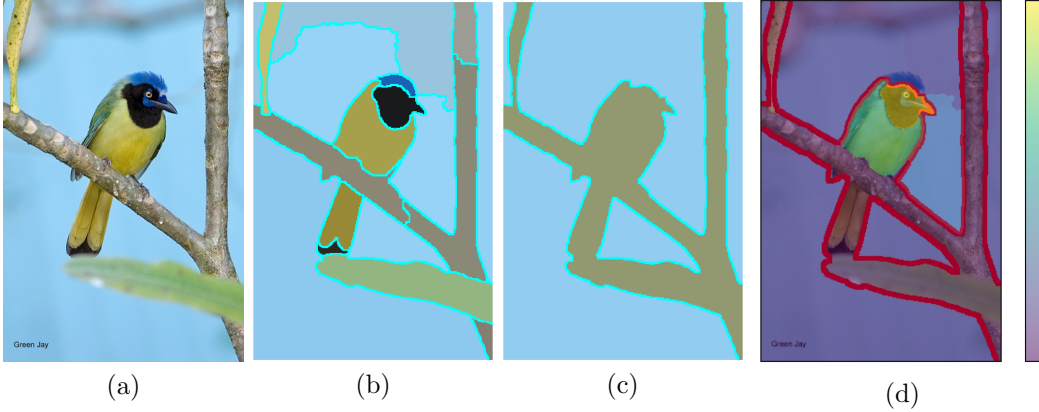


Figure 2: Influence of inaccurate semantic segmentation on Instance Green_Jay_0085_66077 in CUB-200-2011 dataset [2].

As shown in Fig. 1(a) and Fig. 2(a), the two images represent distinct segmentation challenges. The first image shows a seagull standing on a wooden platform in front of grey urban buildings under cloudy weather. The visual similarity between the seagull’s body and the background buildings leads to ambiguous semantic boundaries between the foreground and background regions. The second image presents a green jay standing on a branch, where the bird’s colorful feathers introduce some semantic confusion with the sky and the branch.

For both images, we applied the fuzzy C-means (FCM) [3] clustering algorithm to perform semantic segmentation on the extracted superpixels (Fig. 1(b) and Fig. 2(b)), with the expected number of semantic segments set to two. Because FCM relies primarily on color similarity, the segmentation in Fig. 1(c) and Fig. 2(c) can not meet our desired semantic grouping. Specifically, in Fig. 1(c), the seagull’s body, the sky, and part of the lake are assigned to the light-colored segment, while the seagull’s tail and the background buildings are grouped into the dark-colored segment. In Fig. 2(c), the bird’s body and the tree branch are merged into a green segment, while its head feathers and the sky are categorized

together into a blue segment. These results reflect typical segmentation inaccuracies caused by color similarity and object overlap.

Based on these inaccurate segmentation maps, we trained GT2-HFMAE-I explainers for each case and generated the corresponding explanations of the classifier’s behavior, as shown in Fig. 1(d) and Fig. 2(d). At the superpixel level, the produced explanations remain largely correct: the superpixels corresponding to the seagull and the bird are highlighted with high salience, indicating that the classifier’s decision basis is still captured accurately. This observation supports our theoretical analysis in the previous section: the proposed method can reduce the impact of segmentation uncertainty by fuzzy membership modeling in the explainer.

However, at the semantic level, the influence of segmentation inaccuracy becomes more evident. In Fig. 1(d), the light-colored semantic segment (the seagull, the sky, and part of the water surface) is identified as the explanation for the classifier’s decision, while in Fig. 2(d), it is the green segment (the bird and parts of the branch). These results show that, although the explainer correctly identifies the superpixels influencing the classifier’s output, the semantic level explanation becomes partially redundant or incomplete, for instance, the redundant inclusion of the branch in Fig. 2(d) or the absence of the seagull’s tail in Fig. 1(d). This demonstrates how inaccurate segmentation may affect the precision of explanations, while still preserving the general decision basis due to the method’s robustness design.

(3) Practical solutions with the proposed framework handling inaccurate segmentation

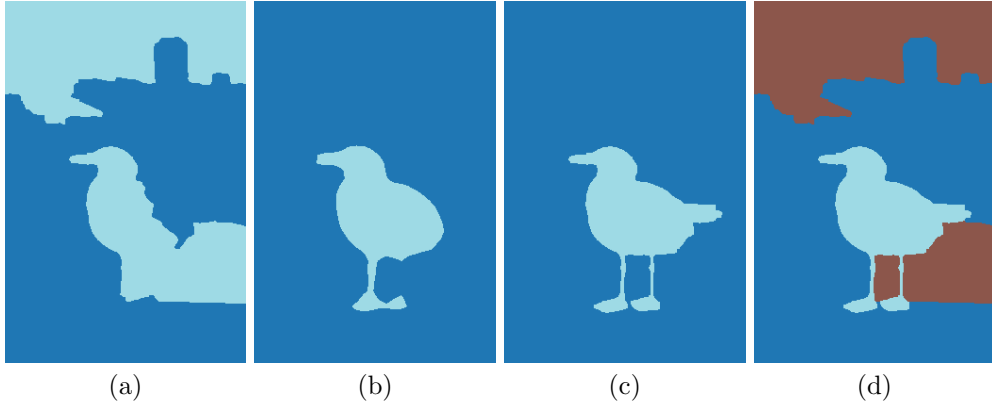


Figure 3: Refinement of semantic segmentation on Instance Western_Gull_0087_54193 in CUB-200-2011 dataset [2].

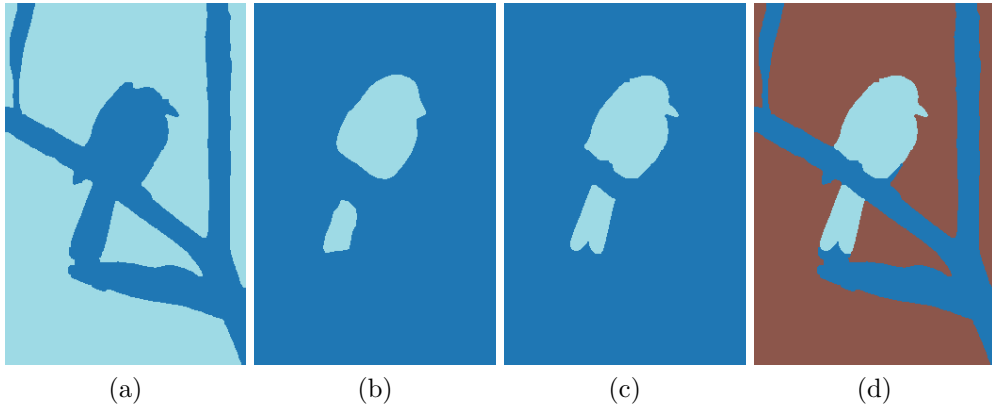


Figure 4: Refinement of semantic segmentation on Instance Green_Jay_0085_66077 in CUB-200-2011 dataset [2].

The semantic axis of the proposed biaxial hierarchical explanation framework provides a practical solution for refining explanations when the initial semantic segmentation is inaccurate. Based on the initial results from FCM shown in Fig. 3(a) and Fig. 4(a), the explanations in the previous subsection revealed that segmentation imprecision mainly affected on the semantic level. To address this issue, users can flexibly utilize the semantic axis to refine the segmentation results by integrating multiple segmentation algorithms, thereby constructing explanations at a finer level.

In this experiment, we combined the FCM segmentation with the output of a pre-trained DeepLabV3 [4] model (provided by the PyTorch library), as illustrated in Fig. 3(b) and Fig. 4(b). DeepLabV3 performs pixel-wise semantic labeling; hence, to align its results with our explainer’s superpixel-based representation, we established a superpixel-to-segment mapping using a majority-voting strategy: for each superpixel, we counted the number of pixels belonging to each semantic segment and assigned the superpixel to the segment with the highest count. The resulting superpixel-level segmentation maps are shown in Fig. 3(c) and Fig. 4(c). Note that DeepLabV3 assigns the Label 0 as the “background” class, which we excluded from subsequent processing.

Finally, we overlaid the non-background semantic segments from DeepLabV3 onto the original FCM segmentation results, obtaining refined semantic maps as shown in Fig. 3(d) and Fig. 4(d). It needs to notice that the PMF in this case needs to be calculated by Eq. 27 and Eq. 28 since the semantic segmentation is verified by the user. Explanations generated based on the refined segmentation are compared in Fig. 5. It can be observed that, after refinement, the explainer captures the classifier’s decision basis with significantly improved semantic accuracy: redundant regions such as branches or backgrounds are eliminated, and the highlighted segments correspond more precisely to the true object of interest.

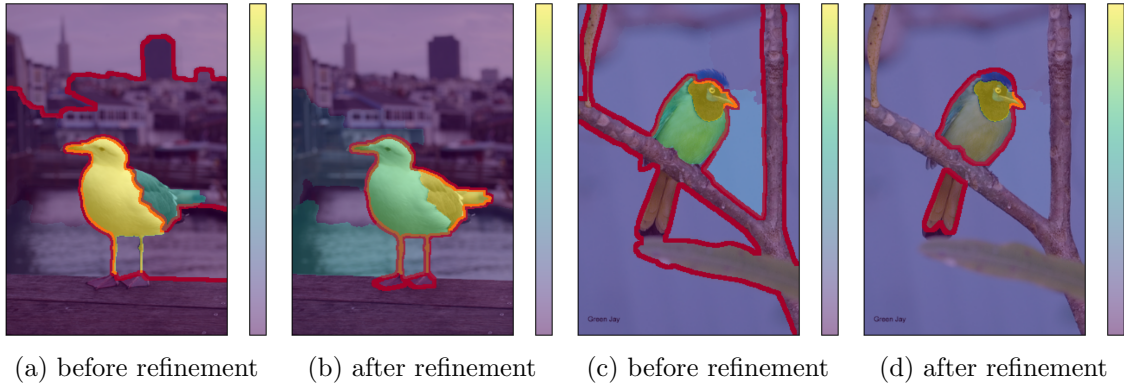


Figure 5: Comparison of explanations before and after refinement of semantic segmentation.

Importantly, throughout this refinement process, no additional explainer training was required. Therefore, the improved interpretability was achieved without introducing extra computational cost beyond the image segmentation.

These analyses and experiments demonstrate that segmentation inaccuracies may moderately degrade the semantic level clarity of the explanations, but such issues can be effectively solved within the existing framework.

References

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